Akhmedkhan (Ahan) Shabanov

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Education

Simon Fraser University — **Vancouver, Canada**
Ph.D. in Computer Science — *Jan. 2023 – Present*
Research on 3D computer vision
Supervisor: **Prof. Andrea Tagliasacchi**

Yandex School of Data Analysis (top 3% of 4000 applicants) — **Moscow, Russia**
Additional master's-level program in CS and ML offered by Yandex — *Sep. 2018 – Jun. 2020*
Featured courses: Self-Driving Cars, Deep (Reinforcement) Learning, Computer Vision, NLP, Algorithms, Crowd Sourcing, Game Theory

Moscow Institute of Physics and Technology — **Moscow, Russia**
M.S. in Informatics and Computer Technology (top 5%, Honour degree) — *Sep. 2019 – Sep. 2021*
GPA: **4.8/5**; Thesis: Monocular 3D head with hair reconstruction
Supervisor: **Dr. Dmitry Ulyanov**

Moscow, Russia
B.S. in Applied Mathematics and Physics — *Sep. 2015 – Aug. 2019*
GPA: **4.5/5**; Thesis: Neural feature descriptors for multi-modal image registration

Research & Work Experience

Meta — **Zurich, Switzerland**
Research Intern — *Sep. 2025 – Present*
- Developing a novel method for generative 3D reconstruction

Huawei — **Canada, Vancouver**
AI Research Intern — *Jan. 2025 – July 2025*
- Led research on controllable generative models as interactive game engines
- Developed a prototype using video diffusion models with game engine-style conditioning
- Investigated consistency and control in synthetic environments via simplified physics simulator

Avaturn (ex. In3D) — **Moscow, Russia**
Computer Vision Research Engineer, Founding team member — *Mar. 2020 – Aug. 2022*
Avaturn - startup building technology for 3D body/head scanning with a smartphone.
- Implemented core pipeline for single/multi-view 3D head and hair reconstruction
- Led research on depth denoising for iPhone TrueDepth sensor (published)
- Developed an automated animation retargeting pipeline for 3D human avatars

European Molecular Biology Laboratory (EMBL) — **Heidelberg, Germany**
Deep Learning Research Intern — *Jul. – Sep. 2019, 2020*
Advisor: Anna Kreshuk
- Analyzed cell nucleoli segmentation in time-lapse microscopy videos
- Developed self-supervised method for temporal consistency in segmentation (published)
- Used in practice by EMBL researchers: significantly reduces annotation workload

Teaching Experience

TA for 3D Computer Vision course — **Vancouver, Canada**
Created and graded assignments for graduate level course — *Spring 2022/2023*
TA for Machine Learning course — **Moscow, Russia**
Conducted weekly tutorial sessions for third and fourth-year undergraduates — *Spring 2020*

Publications & Patents

BANF: Band-limited Neural Fields for Levels of Detail Reconstruction [CVPR 2024]
Ahan Shabanov, Shrisudhan Govindarajan, Cody Reading, Lily Goli, Daniel Rebain, Kwang Moo Yi, Andrea Tagliasacchi

Lagrangian Hashing for Compressed Neural Field Representations [ECCV 2024]
Shrisudhan Govindarajan*, Zeno Sambugaro*, Ahan Shabanov, Towaki Takikawa, Weiwei Sun, Daniel Rebain, Nicola Conci, Kwang Moo Yi, Andrea Tagliasacchi

Methods and systems for generating a resolved threedimensional (R3D) avatar [US Patent 2022]
Dmitry Ulyanov, Igor Pasechnik, Ahan Shabanov, Vadim Lebedev, Ilya Krotov, Nikolay Chinaev, Bulat Yakupov, Vsevolod Poletaev

Self-supervised Depth Denoising Using Lower- and Higher-quality RGB-D sensors [3DV 2020]
Ahan Shabanov, Ilya Krotov, Nikolay Chinaev, Vsevolod Poletaev, Sergei Kozlukov, Igor Pasechnik, Bulat Yakupov, Artsiom Sanakoyeu, Vadim Lebedev, Dmitry Ulyanov

Unsupervised temporal consistency improvement for microscopy video segmentation with Siamese networks [bioRxiv, 2021]
Ahan Shabanov, Daja Schichler, Constantin Pape, Sara Cuylen-Haering, Anna Kreshuk

Optical-to-SAR Image Registration Using a Combination of CNN Descriptors and Cross-Correlation Coefficient [International Conference on Machine Vision (ICMV) 2019, Amsterdam]
Ahan Shabanov, Sergey Gladilin, Evgeny Shvets

A multilayer neural network for determination of particle size distribution in Dynamic Light Scattering problem [Computer Research and Modeling Journal, 2019]
Ahan Shabanov, Mikhail Nikolaevich Petrov, Alexander Viktorovich Chikitkin

Personal/Notable Achievements

Increased State Academic Scholarship *2019, 2020, 2021*
Scholarship for excellent performance and research activity (top 10%, Dean's List)

2nd place at Vision Hackathon **Moscow, Russia**
Hackathon on analyzing videos captured by cameras on driving cars. *2018*
Our team of five won 10k$

2nd place at Junction Hackathon **Helsinki, Finland**
Junction - biggest hackathon in Europe *2018*

1st place in Physics at All-Russian Olympiad **Moscow, Russia**
It was hosted by MIPT. I was granted direct university admission eligibility nationwide *2014*

Skills

- **Programming**: Python, C, C++, Bash, Git, LaTeX
- **ML/CV**: Numpy, Pytorch, TensorFlow, Keras, OpenCV, Kornia
- **3D/Computer Graphics**: Blender, MeshLab, Polyscope, Pytorch3D, libigl, Open3D
- **Languages**: Russian(native), English(C1)